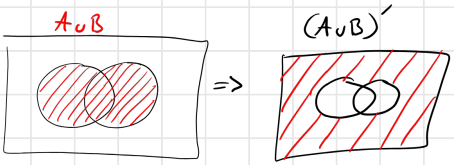
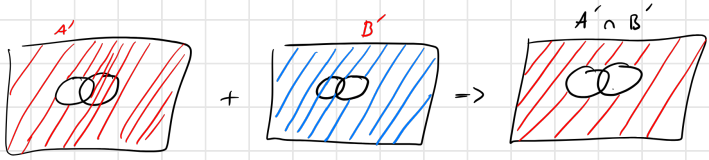
Opg 1

$P(A') = 1 - P(A) \quad \Rightarrow P(A) = 1 - P(A')$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P((A \cup B)')$:



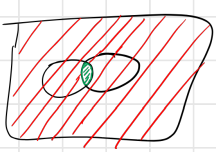


Ser at

$P(A' \cap B') = P((A \cup B)')$

siden $P(A' \cap B')$ er det motsatte/inverse arealet av $P(A' \cup B')$. og tilsvarende for $P((A \cup B)')$ og $P((A \cap B)')$ må $P((A \cap B)') = P(A' \cup B')$ stemme hvis $P(A' \cap B') = P((A \cup B)')$ stemmer

$x = y \Rightarrow x^{-1} = y^{-1}$



$A' \cup B'$

$A \cap B \Rightarrow (A \cap B)'$

Oppg 2

34 kuler, 7 røde, 27 blå

trekker 7 uten tilbakelegg

A: Alle kulene er røde
B: 4 av 7 er røde

$P(A) = \frac{g}{m}$, $m = {}_{34}C_7$ / $g = {}_7C_7$

$P(A) = \frac{1}{\binom{34}{7}} = 0{,}0000000185 = 1{,}85 \cdot 10^{-7}$

$P(B)$: $m = \binom{34}{7}$, g: ${}_7C_4 \cdot {}_{27}C_3$

$P(B) = \frac{g}{m} = 0{,}019 \approx 2\%$

C: etter 1. trekk; én kule blant 27.

C = P for 6 av 7 trekk er rød, deretter en rød
(6 av 7 trekk er rød: uordnet; deretter en rød: ordnet)

$P(C) = \frac{g}{m}$, ${}_7C_6 \cdot {}_{27}C_1 \cdot {}_1P_1$

$m = {}_{34}C_7 \cdot {}_{27}P_1$ = ordnet × uordnet

$$P(C) = \frac{{}_7C_6 \, {}_{27}C_1}{{}_{34}C_7} \cdot \frac{{}_1P_1}{{}_{27}P_1} = 1{,}30 \cdot 10^{-6}$$

Oppg 3)

300 lodd

3 gir A
3 gir B
294 gir ∅

, kjøper 5 lodd

A: én A

$P(A) = \frac{g}{m}$

$\frac{{}_3C_1 \cdot {}_{297}C_4}{{}_{300}C_5} = 0{,}048668$

$= \underline{\underline{0{,}0487}}$

${}_{297}C_4$: ignorerer om Ola trekker gevinst B eller ei

B: minst én A

$P(B) = 1 - P(B')$

B' = ingen A

$P(B'): m = {}_{300}C_5$

$g: {}_{297}C_5$

$P(B) = 1 - \frac{{}_{297}C_5}{{}_{300}C_5}$

$P(B) = \underline{\underline{0{,}0493}}$

$P(C) = 1 - \frac{{}_{294}C_5}{{}_{300}C_5} = \underline{\underline{0{,}0967}}$

Bayes regel: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

visualisering → Gjør om B til utfallsrom, A blir $A \cap B$, hva er $P(A)$?
<→ hva er $P(A|B)$?

Oppg 4

$P(A) = 0,4 \qquad P(C) = 0,3$

$P(B) = 0,3$

$P(A \cup B) = 0,6 \qquad P(A \cup C) = 0,5 \qquad P(B \cup C) = 0,6$

$P(A \cup B \cup C) = 0,7$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(B \cup C) - P(A \cup C) - P(A \cap B) - 2P(A \cap B \cap C)$$

a) $P(A \cap B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= 0,4 + 0,3 - 0,6$

$= 0,1$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{0,1}{0,3}$$

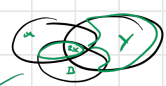
$$= \frac{1}{10} \cdot \frac{10}{3}$$

$$= \frac{1}{3} = 0,33$$

$P(A \cap B \cap C) = P((A \cap B) \cap C)$

$= P(A \cap B) + P(C) - P((A \cap B) \cup C)$

$= 0,1 + 0,6 -$



$P(A \cup B) + P(A \cap C) + P(B \cap C) - P(C) - P(A \cap B) + Y$

$Y = P(A \cup B \cup C) - P(A \cup B)$

$0,4 + 0,3 - 0,5$

$+ \; 0,3 + 0,3 - 0,6$

$- \; 0,3$

$+ \; 0,7 - 0,6$

$= 0,2 - 0,3 + 1 = 0$

$P(A) = 0,4$

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0,1}{0,3} \neq 0,4 \Rightarrow$ De er ikke uavhengige

I tillegg er $P(A \cap B) \neq P(A) \cdot P(B)$

Siden $P(A \cap B) \neq \emptyset$ er de ikke disjunkte

Oppg 5

8% menn og 0,3% kvinner er fargeblinde

Gruppe 1: # kvinner = 2x

menn = x

trekker 1: fargeblind

$$P(\text{kvinne} \mid \text{fargeblind}) = \frac{P(\text{kvinne} \cap \text{fargeblind})}{P(\text{fargeblind})}$$

$P(F)$: F, g: #fargeblinde

$$P(\text{kvinne} \mid \text{fargeblind}) = \frac{P(\text{fargeblind} \mid \text{kvinne})\, P(\text{kvinne})}{P(\text{fargeblind})}$$

$P(\text{fargeblind} \mid \text{kvinne}) = 0{,}003$

$P(\text{kvinne}) = \frac{2}{3} = 0{,}67$ (kvinne 2x menn)

$P(\text{fargeblind}) = 0{,}08 \times \frac{1}{3} + 0{,}003 \times \frac{2}{3}$

$0{,}08x + 0{,}003 \cdot 2x = y$

$x + 2x = 1$

$x = \frac{1}{3}$

$y = 0{,}02841$

$y = P(F) = 0{,}02841$

$P(K \cap F) = \underset{\#\text{kvinner}}{2} \cdot 0{,}67 \cdot 0{,}003$

$$P(\text{kvinne} \mid \text{fargeblind}) = \frac{0{,}003 \cdot 0{,}67}{0{,}02841} = \underline{0{,}0697}$$

$P(K \mid F) = \underline{0{,}0697}$

Oppg 6

$$P(X \leq 2) = F(2) = f(0) + f(1) + f(2) = 0{,}40$$

$$P(X \leq 2 \mid X < 4) = \frac{P(X < 2 \cap X < 4)}{P(X < 4)} = \frac{P(X \leq 2)}{P(X < 4)} = \frac{F(2)}{F(3)} = \frac{0{,}40}{0{,}15 + 0{,}25 + 0{,}40}$$

$$P(X \leq 2 \mid X < 4) = 0{,}50$$

$$P(X \leq 2 \mid X \geq 1) = \frac{P(1 \leq X \leq 2)}{P(X \geq 1)} = \frac{f(1) + f(2)}{\sum_{x=1}^{4} f(x)} = \frac{0{,}35}{0{,}90} = 0{,}388$$

$$F(x) = \sum_{x=0}^{5} f(x)$$

$$F(x) = \{0.05,\ 0.15,\ 0.40,\ 0.80,\ 0.95,\ 1.00\}$$

$$x = \{0,\ 1,\ 2,\ 3,\ 4,\ 5\}$$

$F(x) \in \mathbb{R}$: $F(\lfloor x \rfloor)$ på denne måten vil alle verdier være $\in (\mathbb{R} < 6)$



Eller ønskes et funksjonsuttrykk?

Oppg 7

$$F_X(x) = 1 - e^{-\frac{x^2}{\alpha}} \quad ; \quad x \geq 0 \quad , \alpha > 0$$

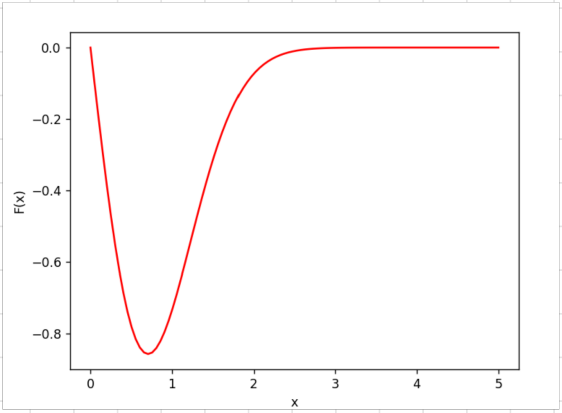
toppunkt: $x = -\sqrt{\frac{\alpha}{2}}$

a)

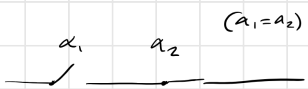
$$F'(x) = f(x) = -\frac{2x}{\alpha} e^{-\frac{x^2}{\alpha}}$$

```python
import numpy as np
import matplotlib.pyplot as plt
# Sett hvilke x-verdier du vil plotte for
x = np.linspace(0,5,100)
# Sett verdien for parameteren alpha
alpha = 1
# Beregn så sannsynlighetstettheten og plott opp funksjonen
def f(x, alpha):
    return -(2*x/alpha)*np.exp(-(x**2)/alpha)

plt.plot(x, f(x, alpha), color="red")
plt.xlabel("x")
plt.ylabel("F(x)")
plt.show()
```



b)



sviktar uavhengig av hverandre

funker så lenge én funker

Z - Levetid

F = % for å leve så lenge $\quad F_Z(z) = P(Z \leq z)$

Z = Levetid
z = slutt på levetid / et tidspunkt

$P(Z \leq z) = P(\text{levetid mindre enn } z)$

$1 - F_Z(z) = P(Z > z)$

$1 - F_Z(z) = P(\text{"instrument fungerer etter } z\text{"})$
$= P(Z > z)$

$1 - F_Z(z) = P(\text{minst én komponent funker etter } z) = P(A')$

$F_Z(z) = P(\text{begge svikter før } z) = P(A)$

$P(A') = 1 - P(A) \Rightarrow P(Z > z) = 1 - P(\text{begge svikter})$

Uavhengighet: $P(A \text{ svikter} \cap B \text{ svikter}) = P(A \text{ svikter}) \cdot P(B \text{ svikter})$ (før z)

$P(Z < z) = F_Z(z)$

$P(A \text{ svikter}) = P(B \text{ svikter})$ pga $\alpha_A = \alpha_B$

$P(Z > z) = 1 - F_Z(z) \cdot F_Z(z) = 1 - F_Z(z)^2$

Oppg 8

300 lodd
3 - A
3 - B
294 - U (ikke gevinst)
Trekker 5

$f_X(x) = P(X=x)$ | punktsannsynlighet

$f_Y(y) = P(Y=y)$

X = # gevinst av A

$f_X(x) = P(x=0) + P(x=1) + \dots$

Y = # gevinst av B

a) $f_X(x) = P(X=x)$

b) $f_{XY}(x,y) = P(X=x, Y=y)$, $x+y \leq 5$

$P(X=1)$: $\frac{{}_3C_1 \cdot {}_{297}C_4}{{}_{300}C_5}$

$f_{XY}(x,y) = \frac{{}_3C_x \cdot {}_3C_y \cdot {}_{294}C_{(5-x-y)}}{{}_{300}C_5}$

$P(X=2)$: $\frac{{}_3C_2 \cdot {}_{297}C_3}{{}_{300}C_5}$

$P(X=x)$: $\frac{{}_3C_x \cdot {}_{297}C_{5-x}}{{}_{300}C_5}$

c) $f_X(x) = \sum_y f_{XY}(x,y)$ | dette skal utledes fra $f_{XY}(x,y)$

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$

$$\sum_{k=0}^{r} {}_mC_k \cdot {}_nC_{(r-k)} = {}_{m+n}C_r$$

$$f_{XY}(x,y) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=0, Y=2) + \dots$$

$x+y \leq 5$

$x, y \in \{0,1,2,3\}$

Skal ha en funksjon av x => Summerer opp verdiene for Y

$$f_{XY}(x,y) = \frac{\binom{3}{x}\binom{3}{y}\binom{294}{5-x-y}}{\binom{300}{5}}$$

Marginal fordeling:

$f_X(x) = \int f_{XY}(x,y)\,dy$

$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)\,dx\,dy = \Sigma_x \Sigma_y f_{XY}(x,y) = 1$, Simultan fordeling

$$\frac{\binom{3}{x}}{\binom{300}{5}} \sum_y \binom{3}{y}\binom{294}{5-x-y} = \frac{\binom{3}{x}}{\binom{300}{5}} \binom{3+294}{5-x}$$

$$= f_X(x)$$

Q.E.D.

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$

$r \leftrightarrow 5-x$

$n \leftrightarrow 294$

$m = 3$

$k = y$